

- 2 (a) Water is the main component of blood.

Explain how the properties of water make it suitable as the main component of blood.

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- (b) Fig. 2.1 is a diagram of the circulation in a mammal.

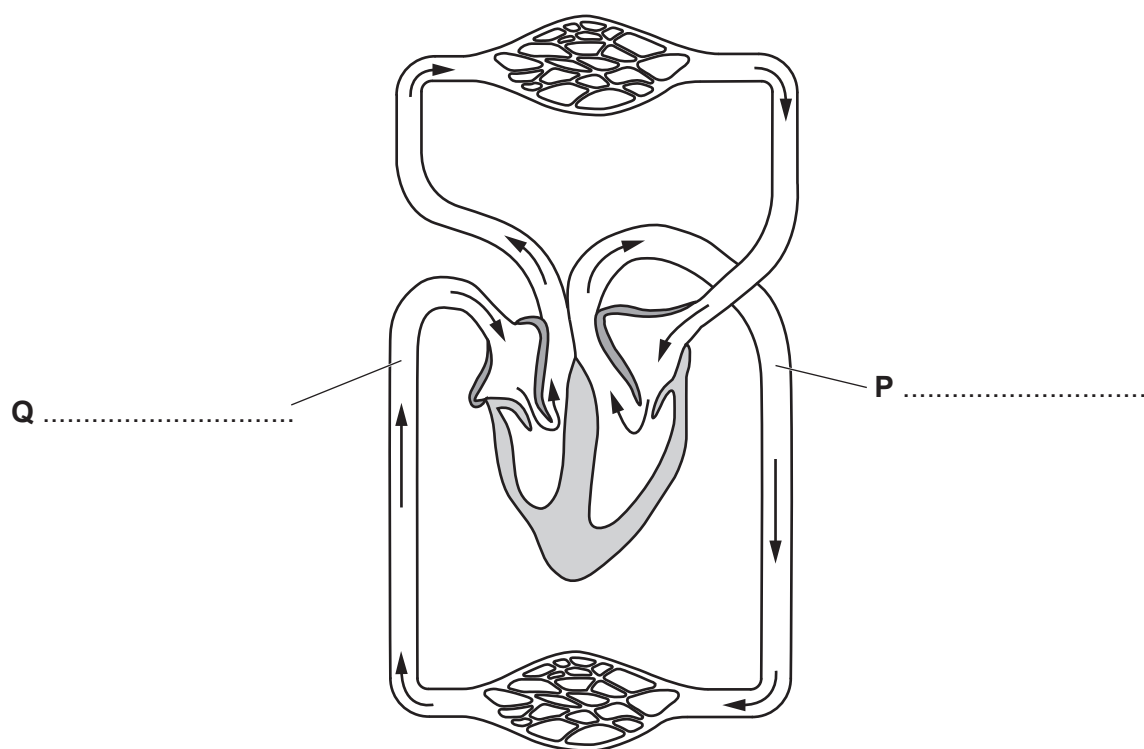


Fig. 2.1

- (i) Complete Fig. 2.1 by naming **P** and **Q**. Write your answers on Fig. 2.1.

[1]

(ii) Describe the functions of **P** and **Q**.

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..... [2]

(iii) Explain why the mammalian circulation is described as a closed, double circulation.

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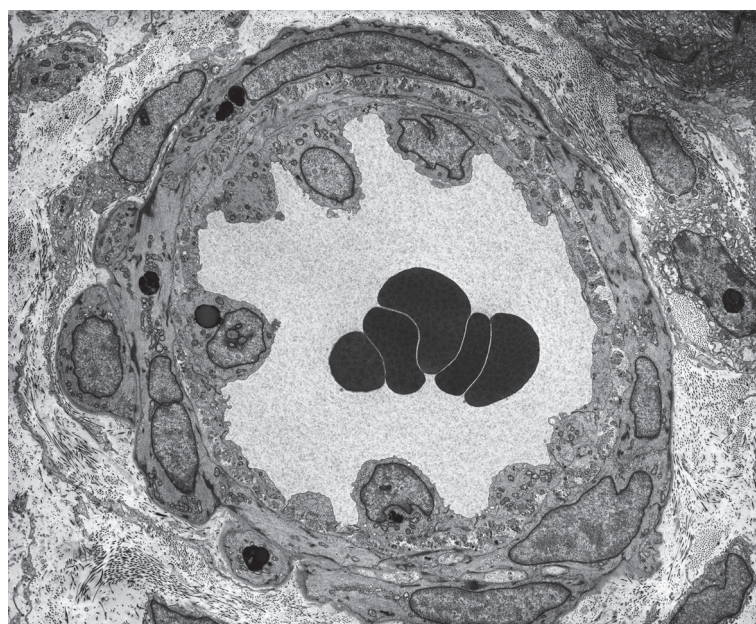
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(c) Fig. 2.2 is a transmission electron micrograph of a cross-section of an arteriole. Blood flows from muscular arteries through arterioles into capillary networks.

The lining of the arteriole is folded because the arteriole has constricted. This constriction causes the blood pressure to decrease from 12.7 kPa in the muscular artery to 2.7 kPa at the end of the arteriole.



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Fig. 2.2

- (i) Explain why it is important that the pressure of blood decreases as it passes through arterioles.

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- (ii) Compare the structure of a muscular artery with the structure of the arteriole shown in Fig. 2.2.

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[Total: 13]